Predicting Home-Based vs. Facility-Based Discharge

A MIMIC-IV self-learning tutorial using early hospitalization data and machine learning

Tutorial goal

- Create a reproducible workflow from raw MIMIC-IV tables to one modeling dataset.
- Compare Logistic Regression, Random Forest, and XGBoost with full vs PCA-reduced lab features.
- Show why threshold selection matters for imbalanced healthcare prediction.

350,699
hospitalizations

163,555
patients

21.3%
facility-based rate

Google Colab Notebook: Self Learning Tutorial.ipynb

Note: Speaker notes are included in this presentation and can be viewed by downloading as a .pptx file.
A separate PDF with slides and notes has also been included.

Tutorial roadmap: eight steps from raw tables to a tested model

Read a slide, then run the corresponding notebook section. The footer on every slide has the current step and where it is in the notebook.

BUILD THE DATASET

1 Set up access

Get MIMIC-IV credentials, link BigQuery, and authenticate the Colab notebook.

Notebook: *Authenticate in Google Cloud (first cells)*

Slide 3

2 Define outcome & cohort

Label discharges home (0) vs facility (1); keep adult survivors with ≥24-hour stays.

Notebook: *Defining the Prediction Outcome · Defining Our Cohort*

Slides 4–5

3 Engineer features

Summarize 9 first-24-hour labs as first / min / max / mean: one row per hospitalization.

Notebook: *Selecting First-24-Hour Laboratory Features*

Slide 6

4 Explore the data

Check class balance, age, admission context, lab missingness, and lab correlations.

Notebook: *Exploratory Data Analysis (EDA)*

Slides 7–10

MODEL AND EVALUATE

5 Split & preprocess

Split by patient; learn clipping, imputation, encoding, scaling, and PCA on subtrain only.

Notebook: *Data Preprocessing and Train/Validation/Test Construction*

Slide 11

6 Train & compare

Fit LogReg, Random Forest, and XGBoost on Full vs PCA features; compare on validation.

Notebook: *Logistic Regression → Compare Model Performance*

Slides 12–13

7 Choose a threshold

Pick the probability cut-off on validation data using a pre-set ≥60% recall goal.

Notebook: *Threshold Sensitivity Analysis*

Slide 14

8 Test once & interpret

Refit on all development patients, test once, then interpret results and limitations.

Notebook: *Final Model Training → Conclusion*

Slides 15–18

Before you start: MIMIC-IV access and notebook setup

Everything runs in the cloud: Google Colab executes Python 3.13 and BigQuery serves MIMIC-IV, so nothing is downloaded locally.

1

Get credentialed access to MIMIC-IV

Create a PhysioNet account, complete credentialing (including the CITI “Data or Specimens Only Research” course), and sign the MIMIC-IV data use agreement.

2

Request BigQuery access

On the MIMIC-IV PhysioNet page, request access via Google BigQuery, then confirm you can see physionet-data › mimiciv_3_1_hosp.

3

Copy the notebook

Open the Colab link (title-slide) and choose File › Save a copy in Drive. Sign in with the Google account that has BigQuery access.

4

Set your project, then run in order

Replace project_id with your own Google Cloud project and run cells top to bottom.

Authentication cell *first code cells*

```
from google.colab import auth
auth.authenticate_user()
from google.cloud import bigquery
project_id = "your-gcp-project-id"  # replace accordingly
client = bigquery.Client(project=project_id)
```

Tables used `physionet-data.mimiciv_3_1_hosp`

`admissions` `patients` `labevents` `d_labitems`

Runtime: Random Forest is the slow step (~3 min per fit in the reference run); Logistic Regression and XGBoost each train in under 30s.

Data use rule

Keep row-level MIMIC data inside BigQuery and Colab. Never paste patient rows into slides, chats, or shared files; share code and aggregate results only.

Clinical question and outcome framing

Predict an observed discharge destination using information available after 24 hours.

- **Research question**
  - Can admission information and first-24-hour labs predict home-based vs facility-based discharge?
- **Home-based discharge**
  - HOME and HOME HEALTH CARE.
- **Facility-based discharge**
  - SKILLED NURSING FACILITY, REHAB, CHRONIC/LTAC, ASSISTED LIVING.
- **Excluded destinations**
  - DIED, HOSPICE, ACUTE HOSPITAL, PSYCH FACILITY, AGAINST ADVICE, OTHER/ambiguous.

In the notebook *cohort query: creating the label*

```
CASE
  WHEN a.discharge_location IN ('HOME', 'HOME HEALTH CARE') THEN 0
  WHEN a.discharge_location IN ('SKILLED NURSING FACILITY', 'REHAB',
       'CHRONIC/LONG TERM ACUTE CARE', 'ASSISTED LIVING') THEN 1
END AS facility_discharge   -- WHERE keeps only these six
```

Outcome label

0 Home-based

Primary residence, with or without home services

1 Facility-based

Discharged to institutional/post-acute setting

Key distinction: this predicts the recorded destination, not whether a new facility placement was clinically required.

The Cohort: large, moderately imbalanced, and patient-linked

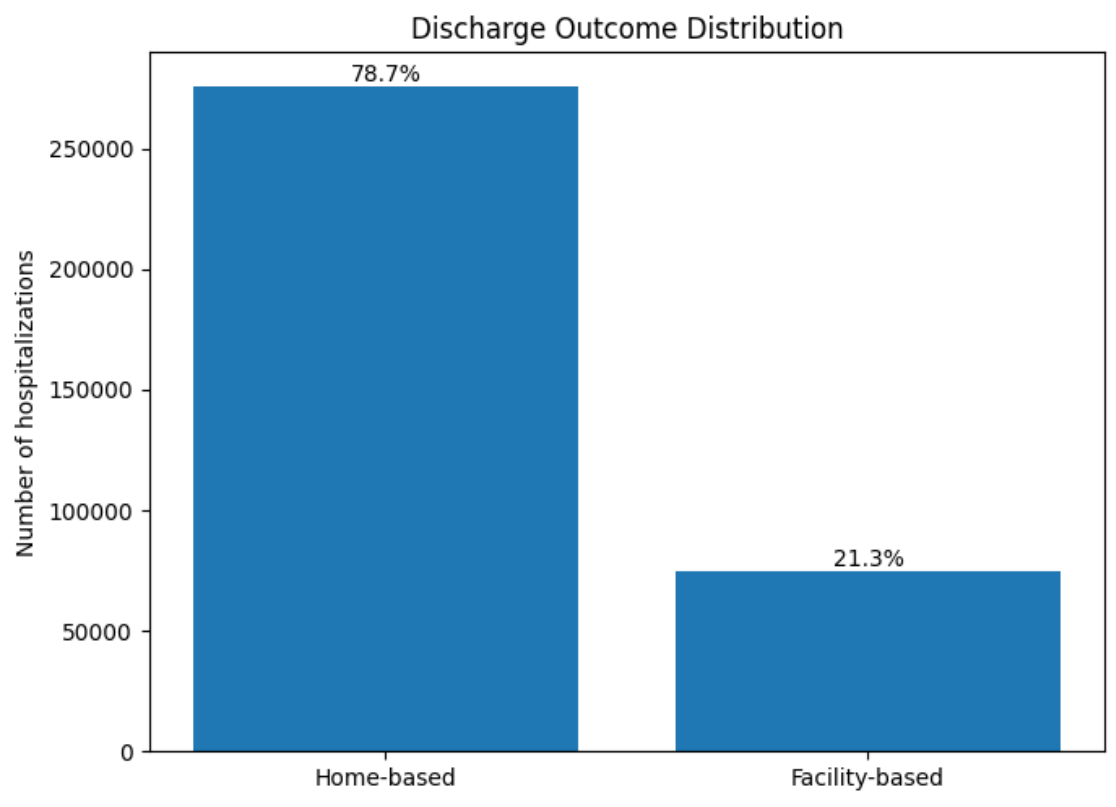


Figure: home-based vs facility-based distribution in eligible admissions.

276,073
home-based admissions

74,626
facility-based admissions

21.3%
positive-class prevalence

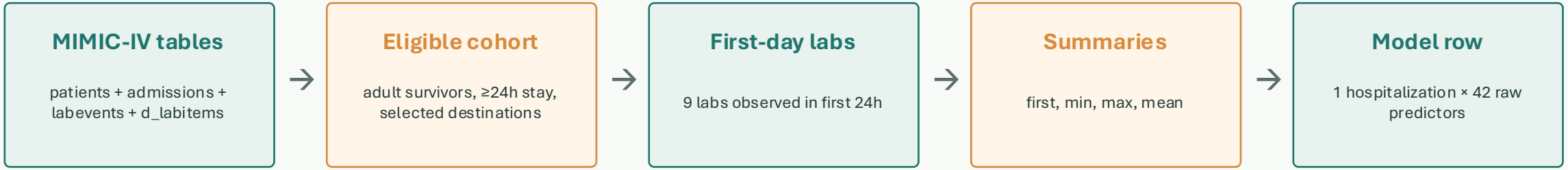
- Each observation = one hospitalization (hadm_id).
- Adult survivors only, with stays lasting at least 24 hours.
- Outcome groups defined before modeling to avoid ambiguity.
- Class imbalance means accuracy alone can be misleading.

Destination composition

Facility category	Admissions
SNF	52,182
Rehab	13,802
Chronic/LTAC	8,064
Assisted living	578

Feature engineering: from raw events to one row per admission

Prediction time = 24 hours after hospital admission; all features must be available by then.



- Lab set: WBC, hemoglobin, platelets, sodium, potassium, creatinine, BUN, glucose, bicarbonate.
- 9 tests × 4 summaries = 36 laboratory features.
  - Summaries: First, min, max, and average values
- Excluded leakage-prone variables: total LOS, discharge summaries, late labs, procedures after 24h, billing diagnoses.

Leakage rule

Only use information that would genuinely be available at the prediction time.

In the notebook *model_sql (abridged): the 24-hour window and the four summaries*

```
WHERE le.charttime >= c.admittime AND le.charttime < c.prediction_time   -- first 24 h only
ARRAY_AGG(valuenum ORDER BY charttime, storetime LIMIT 1)[SAFE_OFFSET(0)] AS first_value,
MIN(valuenum) AS min_value, MAX(valuenum) AS max_value, AVG(valuenum) AS mean_value
GROUP BY hadm_id, itemid   -- then pivot: MAX(IF(itemid = 51301, first_value, NULL)) AS wbc_first, ...
```

EDA: age is strongly associated with discharge destination

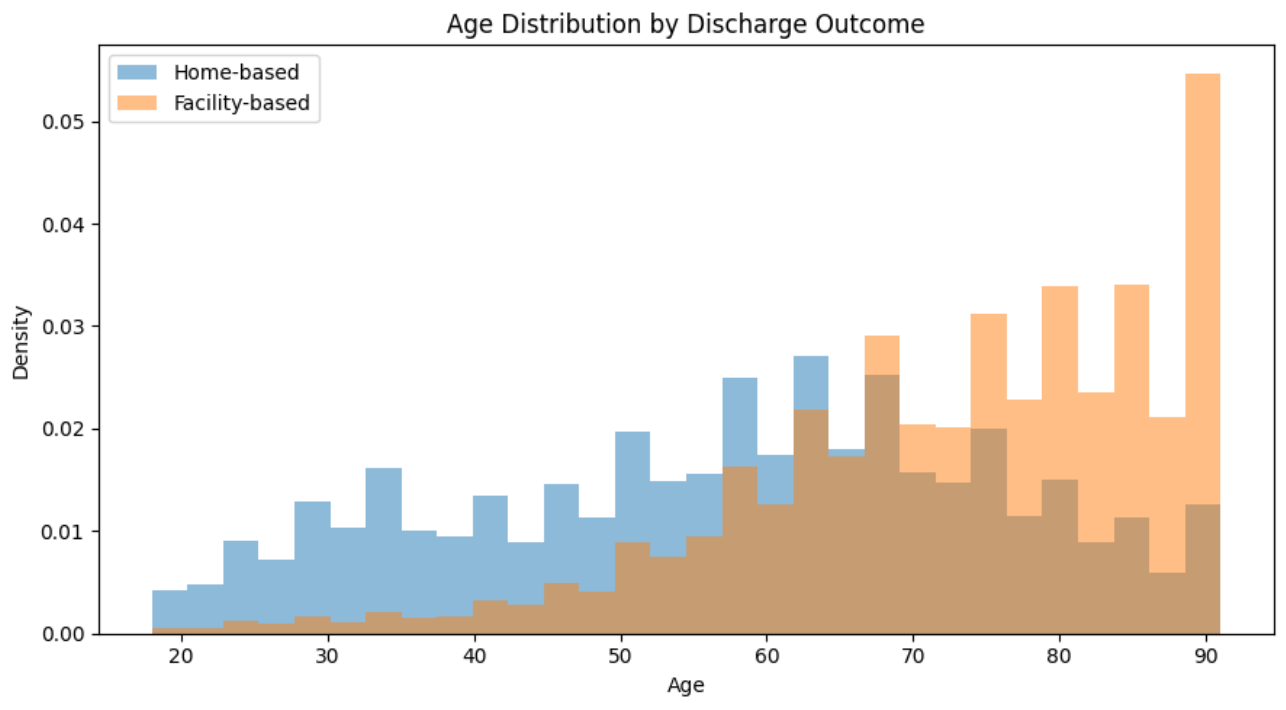


Figure: age distribution by discharge outcome.

57.5
mean age: home-based

72.9
mean age: facility-based

75
median facility age

- Facility-based admissions are substantially older on average.
- Oldest ages are capped/grouped at 91 due to MIMIC de-identification.
- This is an association, not a claim that age alone determines discharge destination.

Group	Mean	Median
Home-based	57.5	59
Facility-based	72.9	75

EDA: admission context carries strong signal

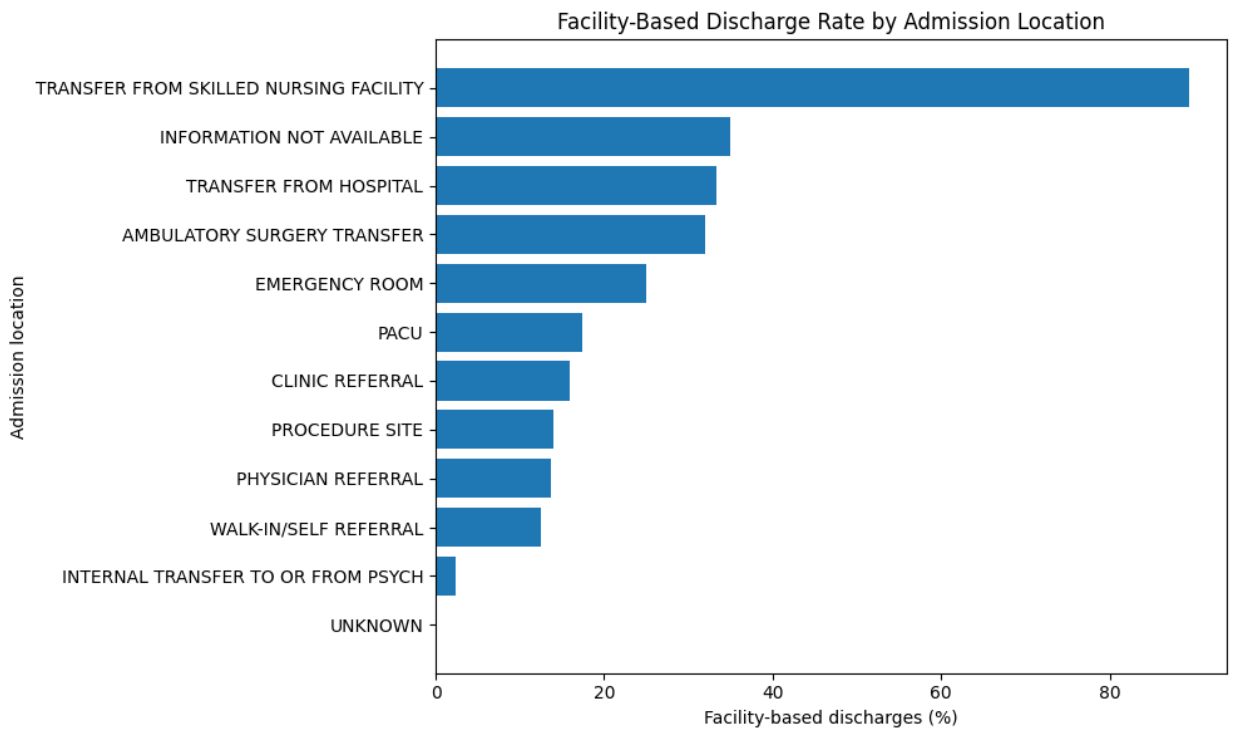


Figure: facility-based discharge rate by admission location; top categories shown.

- Admission location is known at admission and is not considered leakage.
- However, it can act as a proxy for pre-hospital residence or care setting.
- Transfer from SNF: 89.4% facility-based discharge rate.
- Emergency room admissions: 25.1%; physician referral: 13.6%.

Interpretation guardrail

Strong predictive context ≠ causal evidence. A patient admitted from a facility may simply return there.

EDA: laboratory availability is not random

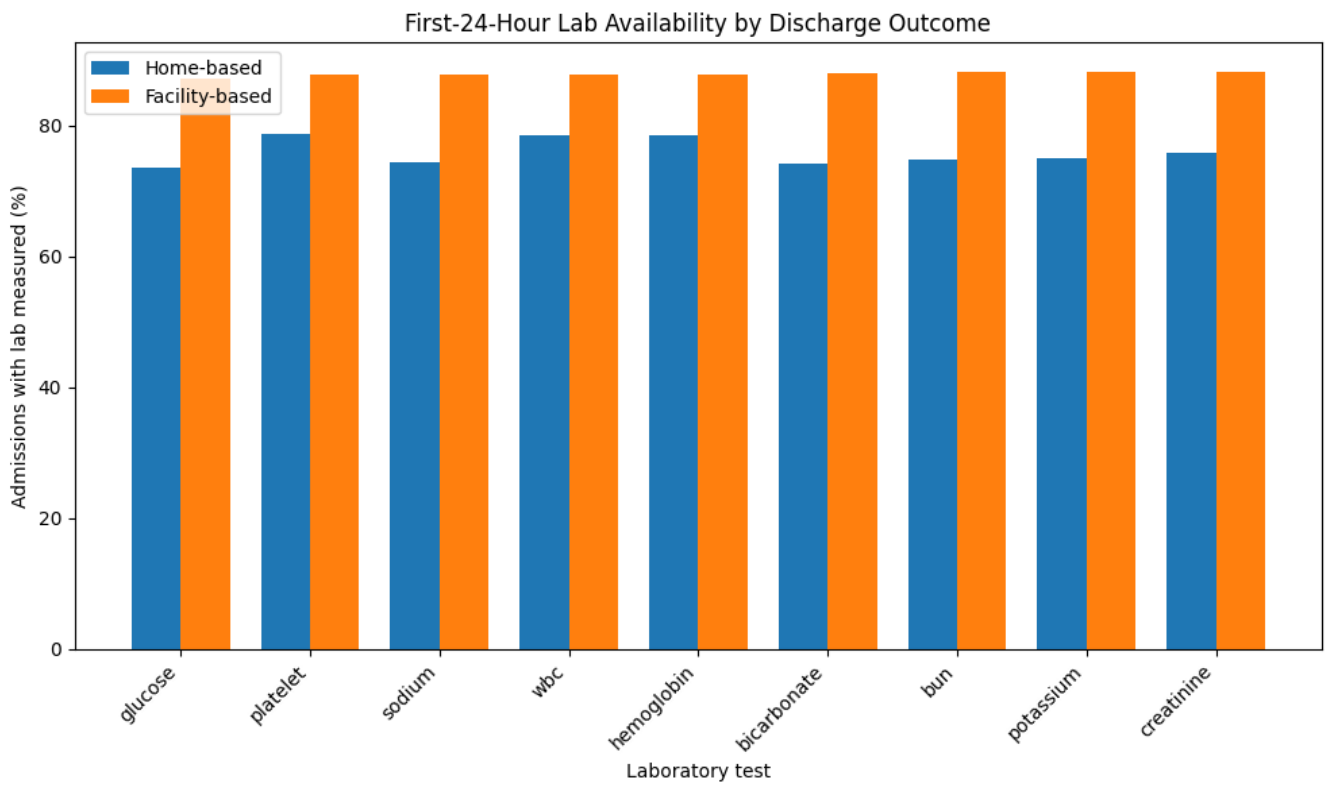


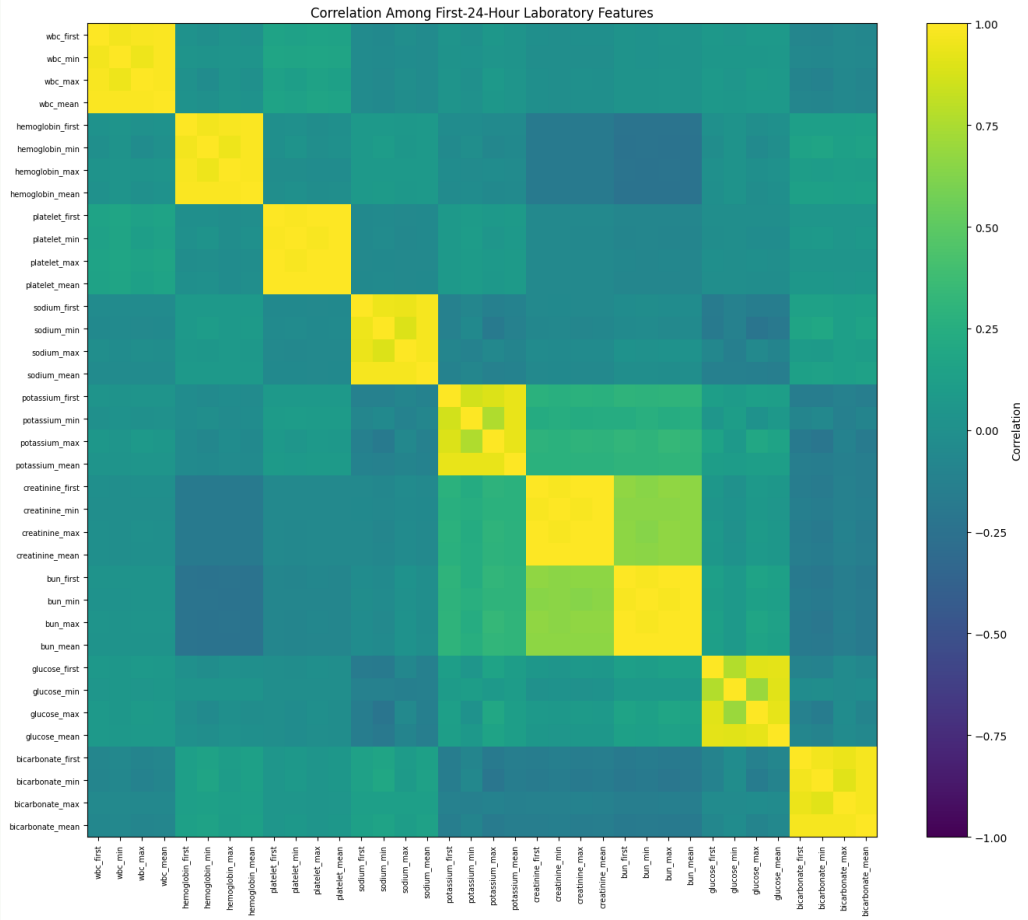
Figure: first-24-hour lab measurement availability by outcome.

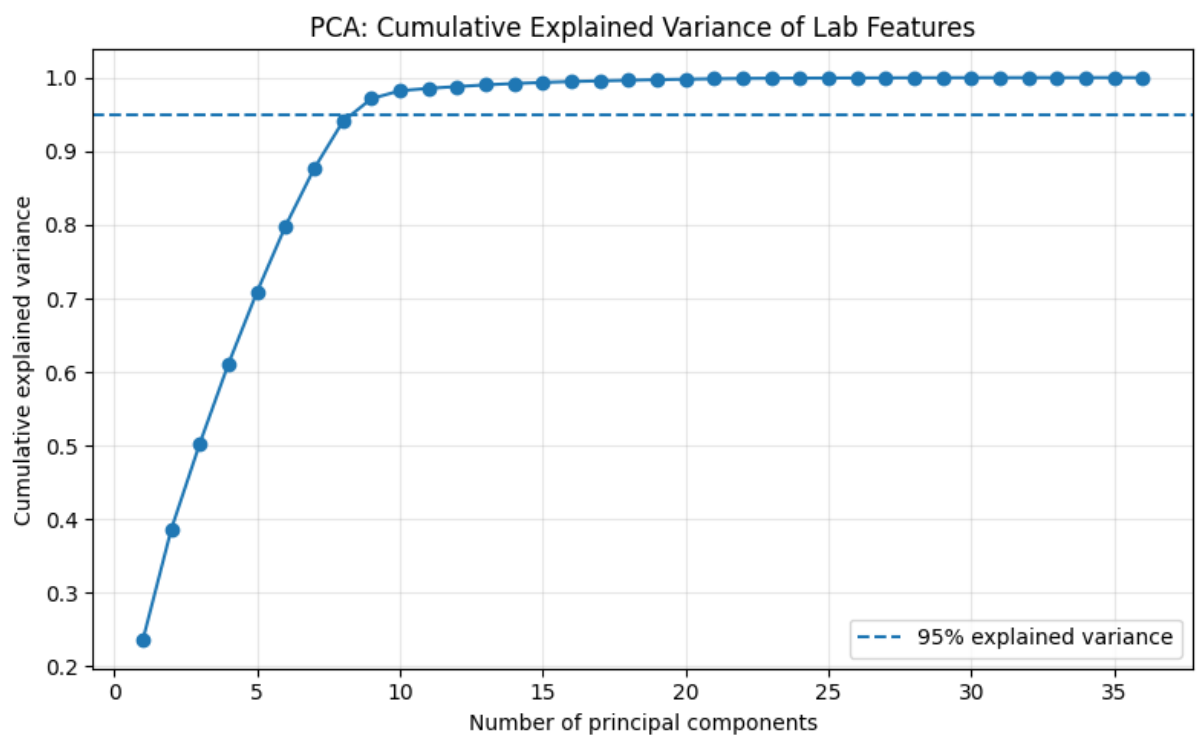
- Facility-based admissions had more complete first-day testing across all nine labs.
- Bicarbonate: 74.3% measured in home-based vs 88.0% in facility-based admissions.
- The gap is about 9–14 percentage points on every lab, largest for bicarbonate, glucose, BUN, and sodium.
- Missingness may reflect clinical complexity or testing intensity.

Modeling choice

Primary models use median imputation; adding lab-missingness indicators is a future extension.

EDA: correlated lab summaries motivate PCA



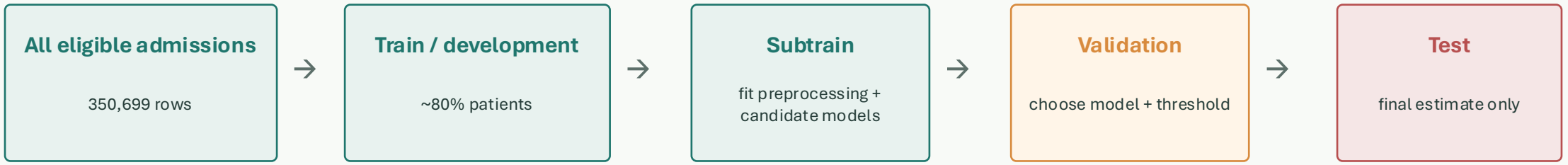


Left: 36-feature lab correlation heatmap. Right: PCA cumulative explained variance.

- Many first/min/max/mean summaries for the same lab are highly correlated.
- Top correlations: BUN first vs max = 0.997; platelets min vs mean = 0.996.
- PCA reduces 36 lab variables to 9 components while retaining ≥95% variance.

Leakage-safe preprocessing and data splitting

All model-selection decisions are made on validation data; test labels remain untouched until final evaluation.



- Split is by subject_id, not by individual admission, to prevent patient-level leakage.
- Subtrain / validation / test outcome balance remains close to 79% home and 21% facility.
- Preprocessing learned from subtrain only: clipping limits, medians, encoder, scaler, PCA.
- After selection, preprocessing is refit on all train/development patients and evaluated once on test.

Split	Home	Facility
Subtrain	78.84%	21.16%
Validation	78.86%	21.14%
Test	78.22%	21.78%

Patient-level split

```
splitter = StratifiedGroupKFold(
    n_splits=5, shuffle=True, random_state=42)
train_idx, test_idx = next(splitter.split(
    model_df, model_df["facility_discharge"],
    groups=model_df["subject_id"]))
```

Fit on subtrain, transform the rest *same pattern for clipping, encoder, scaler, PCA*

```
numeric_imputer.fit_transform(X_subtrain_clipped[numerical_features])  # learn medians
numeric_imputer.transform(X_val_clipped[numerical_features])           # reuse them
```

Modeling experiment: three algorithms × two feature sets

- **Algorithms**
  - Logistic Regression: interpretable linear baseline.
  - Random Forest: nonlinear bagged tree ensemble.
  - XGBoost: sequential boosted-tree classifier.
- **Feature representations**
  - Full: standardized age + 36 lab summaries + encoded admission variables → 70 predictors.
  - PCA: standardized age + 9 lab PCs + encoded admission variables → 43 predictors.

Model	Full	PCA
Logistic Regression	✓	✓
Random Forest	✓	✓
XGBoost	✓	✓

In the notebook *XGBoost-Full; the other five runs follow the same pattern*

```
xgb_full = XGBClassifier(n_estimators=300, max_depth=4, learning_rate=0.05,
                         subsample=0.8, colsample_bytree=0.8,
                         tree_method="hist", random_state=42)
evaluate_model(xgb_full, "XGBoost", "Full",
               X_subtrain_full, X_val_full, y_subtrain, y_val)
# fit on subtrain -> score validation at 0.50 -> log metrics
```

Evaluation metrics

Accuracy, precision, recall, F1, ROC-AUC, PR-AUC (average precision), training time

Why PR-AUC?

Only ~21% of admissions are positive. A no-skill model scores ≈0.21 on PR-AUC but 0.50 on ROC-AUC, so PR-AUC shows minority-class skill more honestly.

Validation results: XGBoost-Full is selected

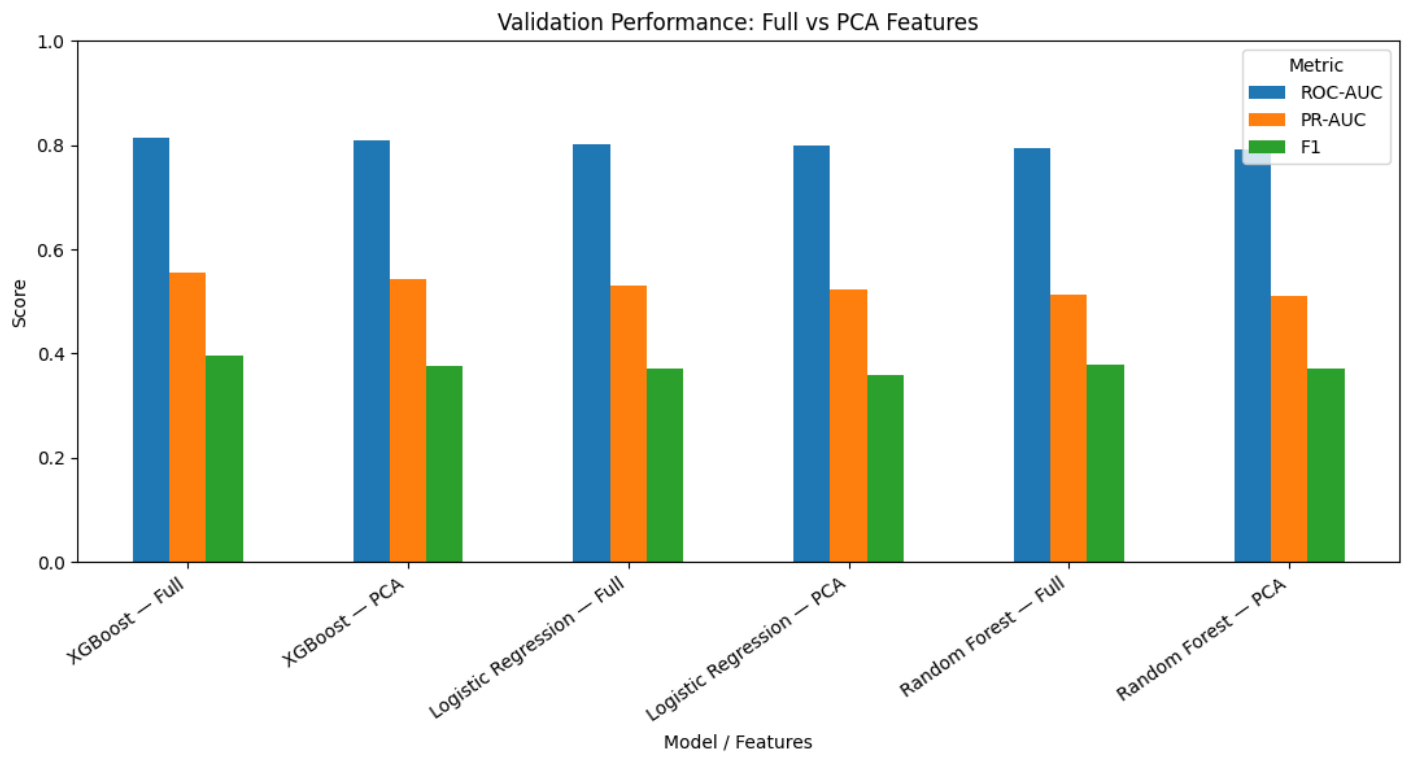


Figure: validation ROC-AUC, PR-AUC, and F1 across all six model configurations.

- XGBoost-Full ranks highest: PR-AUC 0.556 (≈2.6× the 0.21 no-skill baseline), ROC-AUC 0.814.
- Full feature representation slightly outperforms PCA for all three model families.
- PCA preserves variance, but not necessarily all outcome-predictive information.
- Default 0.50 threshold still has low recall: only 28.4% for XGBoost-Full.

Model	Features	PR-AUC	ROC-AUC	Recall @0.50
XGBoost	**Full**	**0.556**	**0.814**	**0.284**
XGBoost	PCA	0.543	0.809	0.266
LogReg	Full	0.531	0.802	0.264
LogReg	PCA	0.524	0.799	0.253
Random Forest	Full	0.515	0.795	0.273
Random Forest	PCA	0.510	0.792	0.270

Threshold selection: turning probabilities into workflow decisions

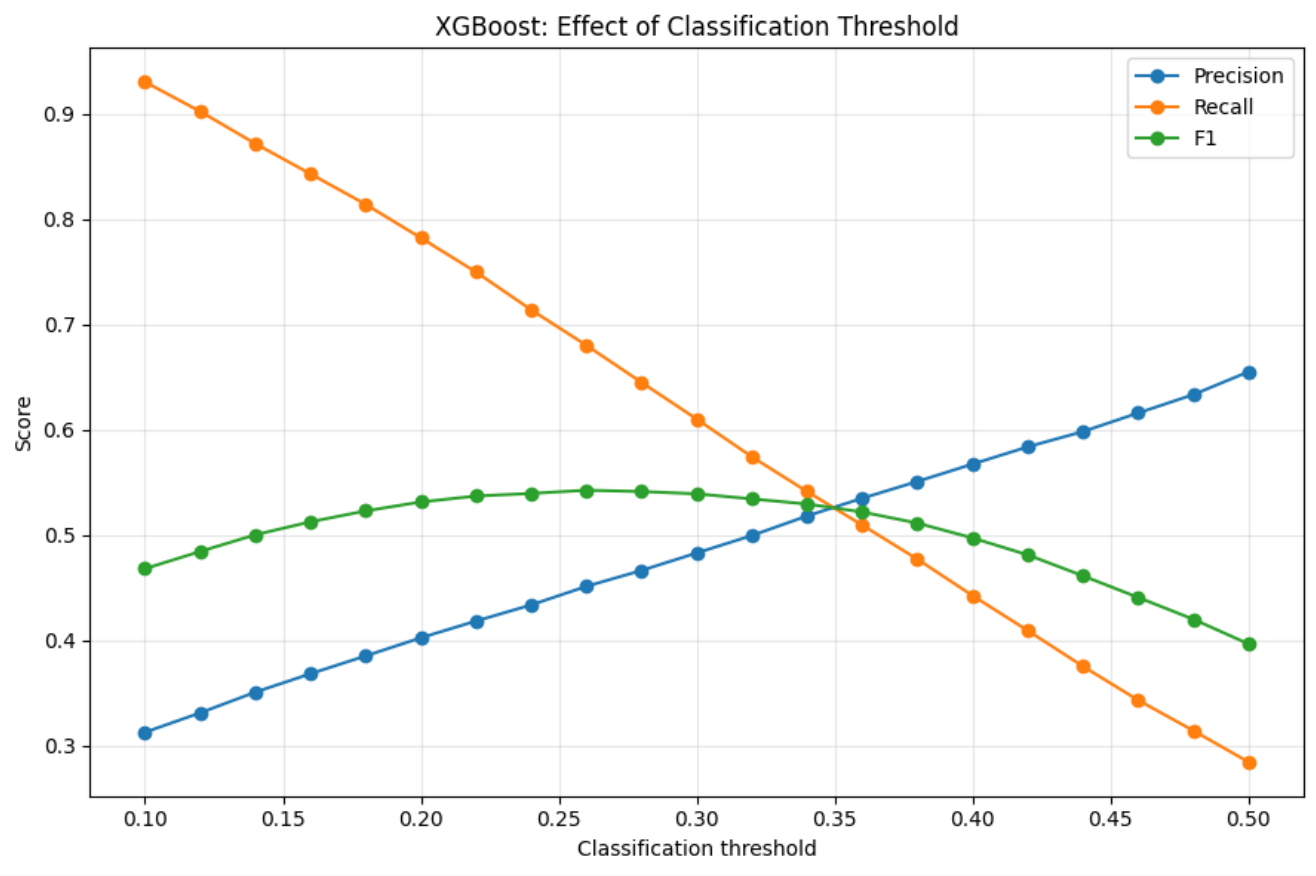


Figure: precision, recall, and F1 for XGBoost-Full across probability thresholds.

- At 0.50, recall is low: 28.4%; only 9.2% of admissions are flagged.
- Lowering the threshold increases recall but reduces precision and specificity.
- Predefined objective: achieve ≥60% recall while keeping threshold as high as possible.
- Validation-selected threshold = 0.30.

Threshold	Prec.	Recall	Specificity	Flagged
0.50	0.654	0.284	0.960	9.2%
0.30	**0.482**	**0.609**	**0.825**	**26.7%**
0.20	0.402	0.781	0.688	41.1%

In the notebook *simplified*

```
# keep thresholds meeting the recall goal; take the highest
eligible = threshold_df[threshold_df["Recall"] >= 0.60]
selected_threshold = eligible["Threshold"].max()   # 0.30
```

Final held-out test evaluation

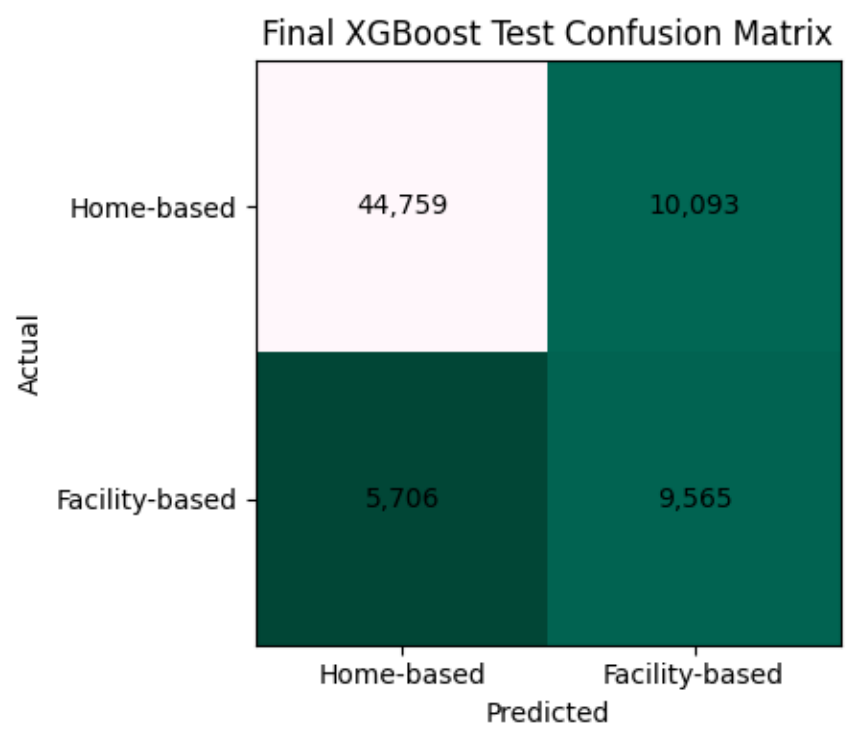


Figure: final XGBoost test confusion matrix at threshold 0.30.

Metric	Value
Accuracy	0.775
Balanced accuracy	0.721
Precision	0.487
Recall	0.626
Specificity	0.816
F1	0.548
ROC-AUC	0.814
PR-AUC	0.564
Flagged	28.0%

- Model identifies 62.6% of facility-based discharges.
- Correctly leaves 81.6% of home-based discharges unflagged.
- Precision 48.7%: about half of flagged admissions truly became facility-based.
- Flagged 28.0%: estimated review workload if used as a screening alert.

Confusion matrix counts: TN 44,759 · FP 10,093 · FN 5,706 · TP 9,565

Why accuracy alone would miss the point

78.2%
accuracy if predicting home for everyone

0%
facility recall for majority baseline

62.6%
facility recall for final model

72.1%
balanced accuracy final model

Threshold	Accuracy	Recall	Precision	Flagged
0.50	0.813	0.291	0.658	9.6%
0.30	0.775	0.626	0.487	28.0%

- Majority baseline looks strong on accuracy but finds no facility-bound patients.
- The selected threshold turns the model into a screening-oriented classifier.
- Use recall, balanced accuracy, F1, PR-AUC, and workload alongside accuracy.

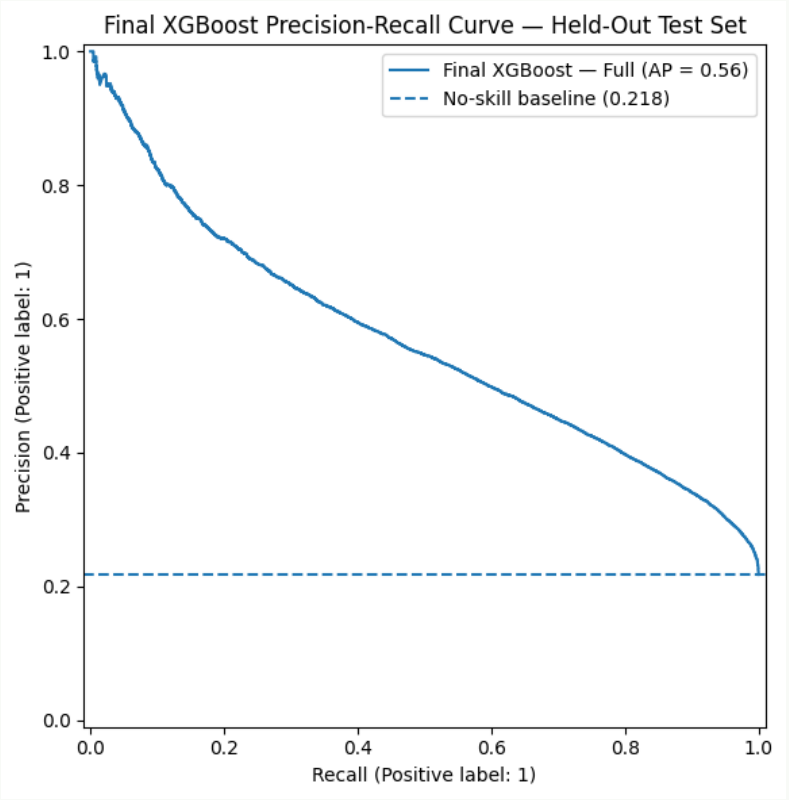


Figure: final test precision-recall curve; no-skill baseline is positive-class prevalence.

Interpreting the selected XGBoost model

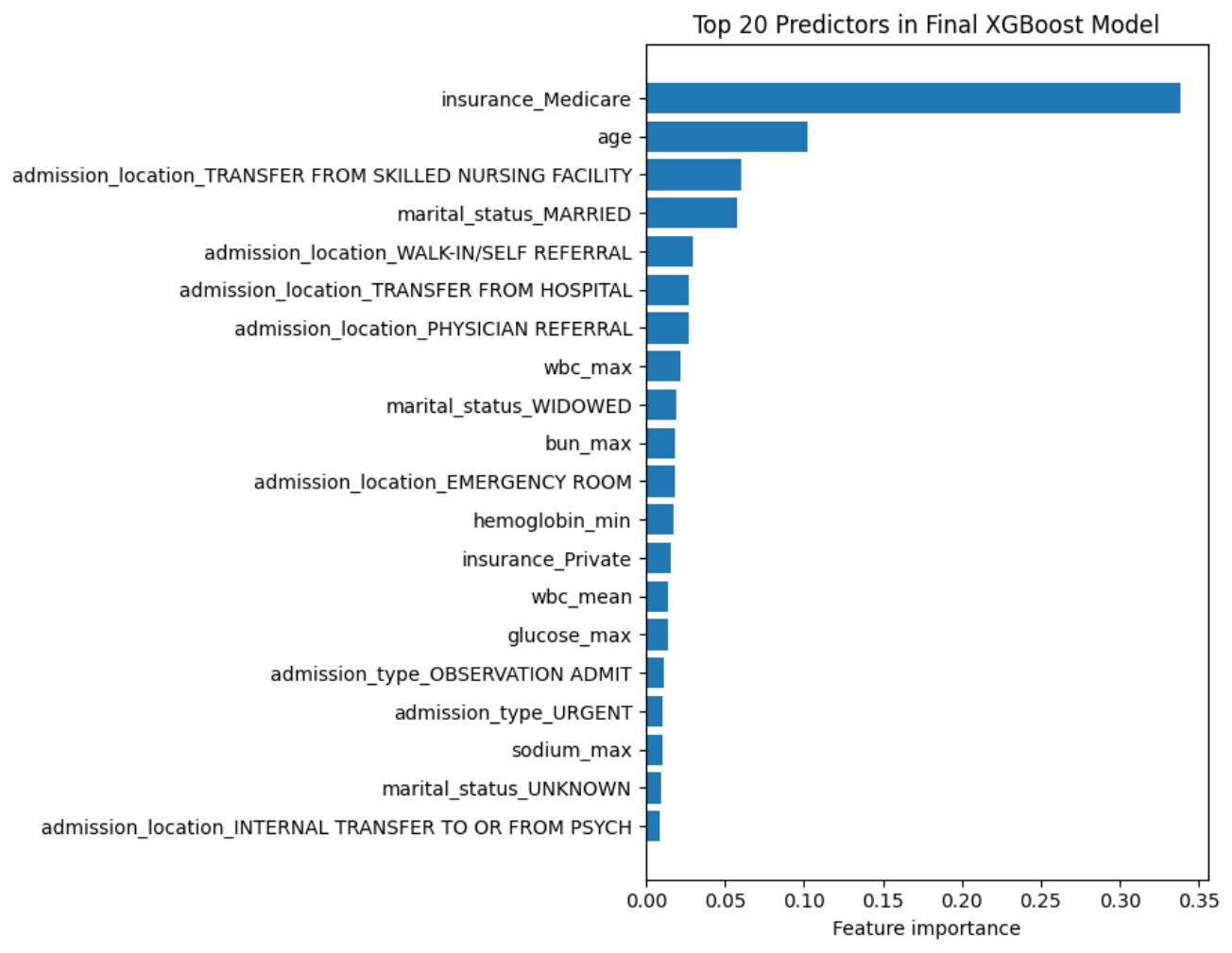


Figure: top 20 feature importances from final refitted XGBoost model.

- insurance_Medicare dominates (~34% of total importance); it partly tracks age ≥65, so correlated features can share credit.
- Admission-from-SNF signal supports the idea that pre-admission setting matters.
- Labs such as WBC, BUN, hemoglobin, glucose, and sodium also appear among top predictors.
- Feature importance is predictive, not causal.

Interpretation warning

Associations used by a model do not prove causes of facility discharge.

Conclusion: limitations and takeaways

- **Limitations**
  - Outcome is recorded destination, not confirmed new need for facility care.
  - Important factors are missing: functional status, mobility, cognition, social support, therapy evaluations.
  - Admission location may proxy for prior residence; lab missingness is informative but not explicitly modeled.
  - Threshold is illustrative; real deployment would require clinical workflow validation and external validation.
- **Takeaways**
  - Early hospitalization data contain meaningful signal for discharge destination.
  - Full lab features slightly outperform PCA across all models.
  - XGBoost-Full performs best on validation and generalizes consistently on test.
  - Threshold choice changes clinical usefulness more than accuracy alone suggests.

Final message

A healthcare ML model is not just an AUC. It is a pipeline that involves defining a clinically meaningful target, preventing leakage, evaluation on patient-disjoint data, choosing an operating threshold, and interpreting tradeoffs.